Ayush Mondal

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Education

RCC Institute of Information Technology (RCCIIT) *2022 – present*
B.Tech in Electronics and Communication Engineering (7.9) *Kolkata, WB*

Purwanchal Vidya Mandir *2020–2022*
Higher Secondary Education/Class 12 th (74%) *Kolkata, WB*

Purwanchal Vidya Mandir *2020*
Secondary Education/ Class 10 th (90%) *Kolkata, WB*

Work Experience

Codec Technologies
Machine Learning Engineer Intern *(Jan'26 - Feb'26)*

- Built a customer churn prediction system using Python, Pandas, NumPy, and Scikit-learn to classify banking customers based on their likelihood of leaving the service.
- Trained a Random Forest classification model using customer demographics, account balance, credit score, tenure, number of products, estimated salary, and account activity.
- Applied feature preprocessing techniques including categorical encoding, standardization, one-hot encoding, and interaction feature creation.
- Engineered behavioral and demographic features such as balance to salary ratio, Product usage, age groups, tenure groups, and geography–gender interactions.
- Achieved 86.95 percent classification accuracy on an 80:20 test split and evaluated churn-specific performance using precision, recall, F1-score, and a confusion matrix.
- Compared Random Forest with Logistic Regression, SVM, and KNN models to identify the best-performing algorithm for the problem.
- Used Joblib model serialization to save and reload the trained classifier and scaler during application inference.
- Implemented a consistent inference pipeline that converts user-entered values into the same feature structure used during model training.

Consistent Services
Software Engineer Intern *(Mar'26 - May'26)*

- Implemented interactive 3D functionality including component selection, zooming, touch/remote-based navigation, and exploded-view visualization, enabling users to inspect individual locomotive parts along with their nomenclature, dimensions, and technical specifications.
- Developed real-time animations and visual simulations for WAP-7 traction and power-flow systems, including the pantograph-to-traction-motor power circuit, traction motor operation, auxiliary motor operation, and regenerative braking workflows.
- Simulated pneumatic and electrical systems, including compressor-to-wheel air flow, pneumatic operations, safety systems, and multi-locomotive electrical, mechanical, and pneumatic coupling processes.
- Created interactive visual training modules covering locomotive safety equipment, including sander, lights, horn, speedometer, vigilance control device, wiper, and hand-brake operations, along with locomotive checking and charging procedures.
- Integrated English and Hindi audio-visual training content and operational descriptions into the visualization system, supporting interactive technical learning and walkthroughs for locomotive personnel.
- Developed the visualization according to WAP-7 technical drawings, specifications, and RDSO/CLW documentation, ensuring equipment nomenclature, dimensions, component representation, and operational workflows matched the provided technical requirements.

Allotrope Systems Private Limited
Backend Engineer Intern *(Jul'25 - Oct'25)*

- Designed and developed scalable RESTful APIs using Node.js and Express.js for Srijoni Healing Home's Hospital Management System, supporting core patient and hospital operational workflows.

- Architected modular backend services for patient records, doctor management, appointments, billing, and diagnostic modules, following MVC architecture and maintainable backend design practices.
- Engineered authentication and role-based authorization mechanisms to enforce access control across administrative and medical user roles, improving security of protected API resources.
- Designed and implemented structured MongoDB schemas using Mongoose with validation and efficient data modeling to maintain data integrity across interconnected hospital workflows.
- Optimized API logic and database queries to improve backend response performance and reliability, while implementing asynchronous request handling and centralized error management.
- Collaborated with frontend developers to integrate REST APIs and deliver end-to-end functionality, while using Git and Postman for version control, API testing, debugging, and development workflows.

Smart India Hackathon (SIH)
Participant *2024*

- Participated in Smart India Hackathon, collaborating with a multidisciplinary team to design and develop a technology-based solution for a real-world problem statement.
- Contributed to solution design, technical development, problem-solving, and team coordination throughout the hackathon.

Projects

Facial Recognition System (Python, OpenCV, CVzone, Computer Vision) GitHub

- Developed a real-time facial recognition attendance system using Python, OpenCV, NumPy, and the face recognition library to automatically identify registered individuals through webcam input.
- Built a face-encoding pipeline that loads registered face images, converts them from BGR to RGB, generates facial embeddings, and stores known encodings for identity matching.
- Implemented real-time face detection and recognition by processing webcam frames, extracting face locations and encodings, comparing them against known embeddings, and selecting the closest match using face-distance calculations.
- Automated attendance logging by recording the recognized individual's name and timestamp into a CSV file while preventing duplicate attendance entries within the existing attendance records.

Self-Driving Car (Python,TensorFlow,OpenCV,Socket.IO) GitHub | Demo Video

- Built an end-to-end behavioural cloning system that trains a convolutional neural network, based on NVIDIA's self-driving architecture, to predict steering angles directly from front-camera images, and deployed it to drive a car autonomously in the Udacity simulator.
- Analysed the driving log with Pandas and Matplotlib, found a heavy bias towards straight driving, and balanced the dataset by capping samples at 400 per steering-angle bin across 25 bins (4,053 to 1,463 samples) to stop the model from learning to only drive straight.
- Used left and right camera images with a ± 0.15 steering correction to teach recovery from off-center positions, expanding the dataset to 4,389 samples, and split it 80:20 into 3,511 training and 878 validation samples.
- Implemented on-the-fly augmentation using imgaug and OpenCV, applying pan ($\pm 10\%$), zoom (up to 1.3x), random brightness, and horizontal flip with steering-angle inversion, each with 50% probability, to improve generalisation across lighting and viewpoints.
- Designed an image preprocessing pipeline (road-region crop, RGB to YUV conversion, Gaussian blur, resize to 200x66, normalisation) and a custom batch generator to stream augmented data during training without loading the full dataset into memory.
- Trained a 5-convolution-layer network with ELU activations and three dropout layers (0.5) using the Adam optimizer and MSE loss, reaching a validation loss of 0.06 over 10 epochs on Google Colab.
- Developed a real-time inference server that receives base64-encoded camera frames and speed telemetry from the simulator over Socket.IO, applies the same preprocessing as training, and returns steering angle and speed-based throttle commands live.

Personal AI Assistant for Recruiter Screening (Python, LangChain, FAISS, Groq) GitHub | Deployed Link

- Built and deployed a recruiter-facing RAG-based AI assistant using Python, LangChain, FAISS, Sentence Transformers, Groq, and Streamlit to answer questions about skills, projects, education, and professional experience.
- Engineered an end-to-end document retrieval pipeline using pypdf for CV text extraction, overlapping text chunking, local Sentence Transformer embeddings, and FAISS vector search for semantic retrieval.

- Implemented top-3 contextual retrieval for each recruiter query and integrated the retrieved CV context with Groq LLM inference to generate grounded, first-person responses while reducing unsupported or hallucinated information.
- Developed a Streamlit web interface and terminal-based chatbot, with configurable chunk size, overlap, retrieval count, and temperature parameters for controlling retrieval quality and response generation.

AWS Data Pipeline (AWS S3, Lambda, Glue, EventBridge, SNS, CloudWatch, Python) GitHub

- Built an event-driven serverless data pipeline on AWS to automate CSV ingestion, transformation, deduplication, and publishing using Amazon S3, AWS Lambda, and AWS Glue.
- Configured Amazon S3 as the raw data source and implemented AWS Lambda-based event orchestration to automatically trigger AWS Glue ETL jobs when new CSV files are uploaded.
- Developed an AWS Glue ETL workflow to read raw CSV data, perform duplicate removal and basic data quality checks, and write cleaned datasets to a curated Amazon S3 location.
- Integrated Amazon EventBridge for pipeline event tracking, Amazon SNS for automated status notifications, and Amazon CloudWatch for logging, monitoring, and operational visibility.

Technical Skills

Languages: Python, SQL, Java
Frameworks/Libraries: Scikit-Learn, Pandas, OpenCV, YOLO, Matplotlib, LangChain, TensorFlow, Express.js, FastAPI
Areas: Computer Vision, NLP, RAG Pipelines, LLM, Generative AI
Tools: Git, GitHub, AWS, VectorDB, Postman, Ollama, Mistral, Gemini API

Certifications & Achievements

- **AMCAT (2025):** Automata – 93/100, Advanced Quant – 84/100, Programming – 79/100
- **AWS Cloud Practitioner** – GeeksforGeeks
- **LangChain: Foundations** – LangChain Academy
- **Intro to Modern AI** – Cisco
- **Intro to Python** – Nasscom